

Unit Testing

Unit testing involves the isolated verification of specific functional components to ensure they meet their design specifications. In this project, unit tests were categorized into Network Integrity, Security Telemetry, and ML Inference Logic.

Test Case 1

Unit Testing	Objectives
Objective	To verify that the virtual lab environment is correctly configured, including VM operation, internal connectivity and network isolation
Action	Boot all virtual machines and perform connectivity and isolation checks using ping and service-level tests
Expected Result	Systems communicate within LABnet, routing functions correctly, and external access is blocked
Actual Result	Internal communication and routing were successful, and external access attempts failed as expected
Conclusion	Test successful

Diagnostics / Ping

Ping

Hostname 192.168.1.102

IP Protocol IPv4

Source address Automatically selected (default)
Select source address for the ping.

Maximum number of pings 3
Select the maximum number of pings.

Seconds between pings 1
Select the number of seconds to wait between pings.



Results

```
PING 192.168.1.102 (192.168.1.102): 56 data bytes
64 bytes from 192.168.1.102: icmp_seq=0 ttl=128 time=10.603 ms
64 bytes from 192.168.1.102: icmp_seq=1 ttl=128 time=2.221 ms
64 bytes from 192.168.1.102: icmp_seq=2 ttl=128 time=2.160 ms

--- 192.168.1.102 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.160/4.995/10.603/3.966 ms
```

Figure 1 pfSense ping to Windows host

Diagnostics / Ping

Ping

Hostname 192.168.1.101

IP Protocol IPv4

Source address Automatically selected (default)
Select source address for the ping.

Maximum number of pings 3
Select the maximum number of pings.

Seconds between pings 1
Select the number of seconds to wait between pings.



Results

```
PING 192.168.1.101 (192.168.1.101): 56 data bytes
64 bytes from 192.168.1.101: icmp_seq=0 ttl=64 time=8.652 ms
64 bytes from 192.168.1.101: icmp_seq=1 ttl=64 time=3.920 ms
64 bytes from 192.168.1.101: icmp_seq=2 ttl=64 time=2.617 ms

--- 192.168.1.101 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.617/5.063/8.652/2.593 ms
```

Figure 2 pfSense ping to Ubuntu server

Ping

Hostname

192.168.57.10

IP Protocol

IPv4

Source address

Automatically selected (default)

Select source address for the ping.

Maximum number of pings

3

Select the maximum number of pings.

Seconds between pings

1

Select the number of seconds to wait between pings.

 Ping

Results

```
PING 192.168.57.10 (192.168.57.10): 56 data bytes
64 bytes from 192.168.57.10: icmp_seq=0 ttl=64 time=13.830 ms
64 bytes from 192.168.57.10: icmp_seq=1 ttl=64 time=1.312 ms
64 bytes from 192.168.57.10: icmp_seq=2 ttl=64 time=1.522 ms

--- 192.168.57.10 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 1.312/5.554/13.830/5.852 ms
```

Figure 3 pfSense ping to Kali Linux

Diagnostics / Ping

Ping

Hostname 192.168.1.100

IP Protocol IPv4

Source address Automatically selected (default)

Select source address for the ping.

Maximum number of pings

3

Select the maximum number of pings.

Seconds between pings

1

Select the number of seconds to wait between pings.



Results

```
PING 192.168.1.100 (192.168.1.100): 56 data bytes
64 bytes from 192.168.1.100: icmp_seq=0 ttl=127 time=24.371 ms
64 bytes from 192.168.1.100: icmp_seq=1 ttl=127 time=24.658 ms
64 bytes from 192.168.1.100: icmp_seq=2 ttl=127 time=61.277 ms

--- 192.168.1.100 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 24.371/36.769/61.277/17.330 ms
```

Figure 4 pfSense ping to Wazuh

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : 
IPv4 Address. . . . . : 192.168.18.41
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.18.1
```

Figure 5 Host machine IP configuration

```

C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vboxuser>ping 192.168.18.41

Pinging 192.168.18.41 with 32 bytes of data:
Reply from 192.168.1.102: Destination host unreachable.
Request timed out.
Reply from 192.168.1.102: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.18.41:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\vboxuser>

```

Figure 6 Windows host unable to reach external network

Test Case 2

Unit Testing	Objectives
Objective	To verify that pfSense monitors and captures network traffic within the lab environment
Action	Generate network traffic to the Wazuh manager (port 1515) and capture packets on pfSense using tcpdump
Expected Result	Network traffic is visible and captured by pfSense, confirming it is correctly positioned as the network gateway
Actual Result	Packet capture on pfSense showed traffic between the Ubuntu system and Wazuh manager on port 1515, confirming successful monitoring
Conclusion	Test successful

```

evani@ubuntu-server:~$ nc -zv 192.168.1.100 1515
Connection to 192.168.1.100 1515 port [tcp/*] succeeded!
evani@ubuntu-server:~$

```

Figure 7 Connectivity Verification (Port 1515)

```

[2.6.0-RELEASE][root@pfSense.home.arpal/root: tcpdump -i em1 port 1515
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on em1, link-type EN10MB (Ethernet), capture size 262144 bytes
01:42:11.376418 IP 192.168.1.101.39290 > 192.168.1.100.1515: Flags [S], seq 810439065, win 64240, options [mss 1460,sackOK,TS val 1551315531 ecr 0,nop,wscale 7], length 0
01:42:11.403596 IP 192.168.1.100.1515 > 192.168.1.101.39290: Flags [S.], seq 1972949109, ack 810439066, win 65160, options [mss 1460,sackOK,TS val 3242529924 ecr 1551315531,nop,wscale 7], length 0
01:42:11.403621 IP 192.168.1.101.39290 > 192.168.1.100.1515: Flags [.], ack 1, win 502, options [nop,nop,TS val 1551315562 ecr 3242529924], length 0
01:42:11.412627 IP 192.168.1.101.39290 > 192.168.1.100.1515: Flags [F.], seq 1, ack 1, win 502, options [nop,nop,TS val 1551315570 ecr 3242529924], length 0
01:42:11.503244 IP 192.168.1.100.1515 > 192.168.1.101.39290: Flags [.], ack 2, win 510, options [nop,nop,TS val 3242530036 ecr 1551315570], length 0
01:42:11.614654 IP 192.168.1.100.1515 > 192.168.1.101.39290: Flags [P.], seq 1:8, ack 2, win 510, options [nop,nop,TS val 3242530153 ecr 1551315570], length 7
01:42:11.614660 IP 192.168.1.101.39290 > 192.168.1.100.1515: Flags [R], seq 810439067, win 0, length 0

```

Figure 8 pfSense Traffic Monitoring (Packet Capture)

Test Case 3

Unit Testing	Objectives
Objective	To verify that attacker traffic from a separate subnet reaches pfSense and is processed according to firewall rules
Action	Generate SYN traffic from Kali Linux using hping3 and capture packets on pfSense
Expected Result	SYN packets from the attacker are observed on pfSense, while the connection attempt is blocked
Actual Result	SYN packets from 192.168.57.10 to 192.168.1.102 were captured on pfSense, while Kali reported 100% packet loss, confirming firewall enforcement
Conclusion	Test successful

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ch:7e:f5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.57.10/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::de3e:3809:bb98:9a32/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ip route
default via 192.168.57.1 dev eth0
192.168.57.0/24 dev eth0 proto kernel scope link src 192.168.57.10

(kali@kali)-[~]
$
```

Figure 9 Kali network configuration showing attacker IP and pfSense gateway.

```
(kali@kali)-[~]
$ sudo hping3 -S 192.168.1.102 -p 445 -c 3
HPING 192.168.1.102 (eth0 192.168.1.102): S set, 40 headers + 0 data bytes

--- 192.168.1.102 hping statistic ---
3 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(kali@kali)-[~]
$
```

Figure 10 SYN attack generated from Kali with connection blocked (100% packet loss).

```
[2.6.0-RELEASE][admin@pfSense.home.arpa]/root: tcpdump -i em3 -nn
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on em3, link-type EN10MB (Ethernet), capture size 262144 bytes
08:38:06.455657 IP 192.168.57.10.2290 > 192.168.1.102.445: Flags [S], seq 212625544, win 512, length 0
08:38:07.405914 IP 192.168.57.10.2291 > 192.168.1.102.445: Flags [S], seq 2050679605, win 512, length 0
08:38:08.360475 IP 192.168.57.10.2292 > 192.168.1.102.445: Flags [S], seq 2024254991, win 512, length 0
08:38:11.215655 ARP, Request who-has 192.168.57.1 tell 192.168.57.10, length 46
08:38:11.218676 ARP, Reply 192.168.57.1 is-at 08:00:27:75:d0:84, length 28
```

Figure 11 pfSense capture showing attacker SYN packets reaching the firewall

Test Case 4

Unit Testing	Objectives
Objective	To verify that the Wazuh agent is correctly installed on the Windows endpoint and successfully connected to the Wazuh manager
Action	Install and start the Wazuh agent on the Windows machine and verify its status from the Wazuh manager while generating endpoint activity
Expected Result	The agent appears as active in the Wazuh manager and logs from the Windows endpoint are received
Actual Result	The agent was successfully installed and appeared as active in the Wazuh manager, and security events generated on the Windows endpoint were received
Conclusion	Test successful

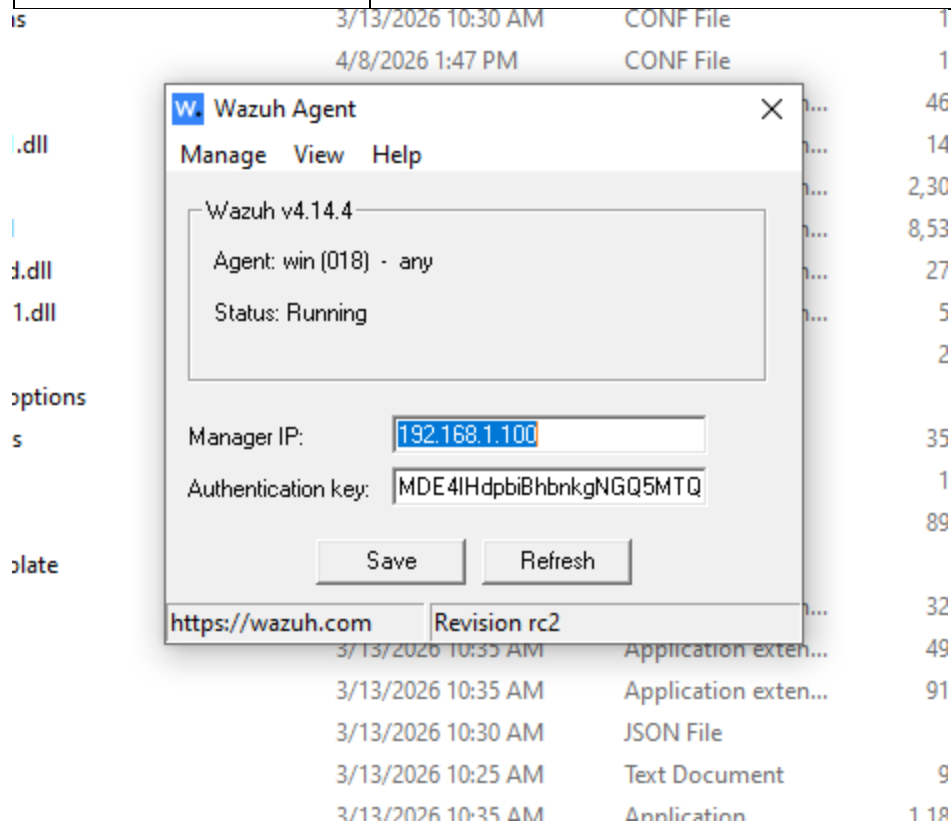


Figure 12 Wazuh agent running on Windows endpoint and connected to manager (192.168.1.100)


```
[wazuh-user@wazuh-server ~]$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
  ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
  ID: 018, Name: win, IP: any, Active
```

Figure 13 Wazuh agent status showing Windows endpoint as active

```
PS C:\Users\vboxuser> runas /user:Administrator "cmd.exe"
Enter the password for Administrator:
Attempting to start cmd.exe as user "WIN\Administrator" ...
PS C:\Users\vboxuser>
```

Figure 14 Command execution on Windows endpoint used to generate security activity

```
{
  "processName": "C:\\Windows\\System32\\svchost.exe", "ipAddress": ":::1", "ipPort": "0", "impersonationLevel": "%1833", "virtualAccount": "%1843", "targetLinkedLogonId": "0x0", "elevatedToken": "%1842"}},
  "location": "EventChannel"}
{
  "timestamp": "2026-04-09T02:56:47.194+0000", "rule": {
    "level": 3, "description": "Special privileges assigned to new logon.", "id": "67028", "mitre": {
      "id": ["T1484"], "tactic": ["Defense Evasion", "Privilege Escalation"], "technique": ["Domain Policy Modification"]}, "firedtimes": 3, "mail": false, "groups": ["windows", "WEF"]}, "agent": {
    "id": "018", "name": "win", "ip": "192.168.1.102"}, "manager": {
    "name": "wazuh-server"}, "id": "1775703407.2061306", "decoder": {
    "name": "windows_eventchannel"}, "data": {
    "win": {
      "system": {
        "providerName": "Microsoft-Windows-Security-Auditing", "providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}", "eventID": "4672", "version": "0", "level": "0", "task": "12548", "opcode": "0", "keywords": "0x8020000000000000", "systemTime": "2026-04-09T02:11:19.4810684Z", "eventRecordID": "15697", "processID": "652", "threadID": "6612", "channel": "Security", "computer": "win", "severityValue": "AUDIT_SUCCESS", "message": "\Special privileges assigned to new logon.\r\n\r\nSubject:\r\n\tSecurity ID:\t\tS-1-5-21-2258376973-2974719183-2099427032-500\r\n\tAccount Name:\t\tAdministrator\r\n\tAccount Domain:\t\tWIN\r\n\tLogon ID:\t\t0x44864f\r\n\tPrivileges:\t\tSeSecurityPrivilege\r\n\t\tSeTakeOwnershipPrivilege\r\n\t\tSeLoadDriverPrivilege\r\n\t\tSeBackupPrivilege\r\n\t\tSeRestorePrivilege\r\n\t\tSeDebugPrivilege\r\n\t\tSeSystemEnvironmentPrivilege\r\n\t\tSeImpersonatePrivilege\r\n\t\tSeDelegateSessionUserImpersonatePrivilege"}}, "eventdata": {
      "subjectUserSid": "S-1-5-21-2258376973-2974719183-2099427032-500", "subjectUserName": "Administrator", "subjectDomainName": "WIN", "subjectLogonId": "0x44864f", "privilegeList": "SeSecurityPrivilege SeTakeOwnershipPrivilege SeLoadDriverPrivilege SeBackupPrivilege SeRestorePrivilege SeDebugPrivilege SeSystemEnvironmentPrivilege SeImpersonatePrivilege SeDelegateSessionUserImpersonatePrivilege"}}, "location": "EventChannel"}
```

Figure 15 Security event received from Windows agent (ID 018) in Wazuh logs

Test Case 5

Unit Testing	Objectives
Objective	To verify that Sysmon security events are generated on the Windows endpoint and successfully ingested and processed by the Wazuh manager

Action	Monitor Sysmon-related activity from the Windows endpoint using Wazuh terminal logs
Expected Result	Sysmon events are captured and processed into alerts within the Wazuh monitoring system
Actual Result	Sysmon Event ID 11 (file creation activity) was captured from the Windows endpoint and processed into a Wazuh alert with associated rule and MITRE classification
Conclusion	Test successful

```

Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd c:\program files\sysmon

c:\Program Files\Sysmon>sysmon64.exe -accepteula -i sysmonconfig-export.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.

```

Figure 16 Sysmon Status

```

[wazuh-user@wazuh-server ~]$ sudo tail -f /var/ossec/logs/alerts/alerts.json | grep -i sysmon
{"timestamp":"2026-04-19T11:28:26.153+0000","rule":{"level":15,"description":"Executable file dropped in folder commonly used by malware","id":"92213","mitre":{"id":["T1105"],"tactic":["Command and Control"],"technique":["Ingress Tool Transfer"]},"firedtimes":2,"mail":true,"groups":["sysmon","sysmon_eid11_detections","windows"]},"agent":{"id":"018","name":"win","ip":"192.168.1.102"},"manager":{"name":"wazuh-server"},"id":"1776598106.371190","decoder":{"name":"windows_eventchannel"},"data":{"win":{"system":{"providerName":"Microsoft-Windows-Sysmon","providerGuid":"{5770385f-c22a-43e0-bf4c-06f5698ffb9d}","eventId":"11","version":"2","level":"4","task":"11","opcode":"0","keywords":"0x8000000000000000","systemTime":"2026-04-19T08:52:15.2587913Z","eventRecordID":"18102","processID":"2944","threadID":"3664","channel":"Microsoft-Windows-Sysmon/Operational","computer":"win","severityValue":"INFORMATION","message":"\\\"File created:\\r\\nRuleName: EXE\\r\\nUtcTime: 2026-04-19 08:52:15.220\\r\\nProcessGuid: {b4a42fe7-942a-69e4-d900-000000001200}\\r\\nProcessId: 8356\\r\\nImage: C:\\\\Program Files (x86)\\\\Microsoft\\\\EdgeUpdate\\\\MicrosoftEdgeUpdate.exe\\r\\nTargetFilename: C:\\\\Users\\\\vboxuser\\\\AppData\\\\Local\\\\Temp\\\\{51F8EC9A-2E29-4822-938A-8EFF67AAE4E9}-MicrosoftEdge_X64_147.0.3912.72_147.0.3912.60.exe\\r\\nCreationUtcTime: 2026-04-19 08:52:15.220\\r\\nUser: NT AUTHORITY\\\\\\\\SYSTEM\\\\\\\\\"}},\"eventdata\":{\"ruleName\":\"EXE\",\"utcTime\":\"2026-04-19 08:52:15.220\",\"processGuid\":\"{b4a42fe7-942a-69e4-d900-000000001200}\",\"processId\":\"8356\",\"image\":\"C:\\\\Program Files (x86)\\\\\\\\Microsoft\\\\\\\\EdgeUpdate\\\\\\\\MicrosoftEdgeUpdate.exe\",\"targetFilename\":\"C:\\\\Users\\\\\\\\vboxuser\\\\\\\\AppData\\\\\\\\Local\\\\\\\\Temp\\\\\\\\{51F8EC9A-2E29-4822-938A-8EFF67AAE4E9}-MicrosoftEdge_X64_147.0.3912.72_147.0.3912.60.exe\",\"creationUtcTime\":\"2026-04-19 08:52:15.220\",\"user\":\"NT AUTHORITY\\\\\\\\\\\\\\\\SYSTEM\\\\\\\\\"}},\"location\":\"EventChannel\"}

```

Figure 17 Sysmon Event ID 11 (file creation) captured

Test Case 6

Unit Testing	Objectives
---------------------	-------------------

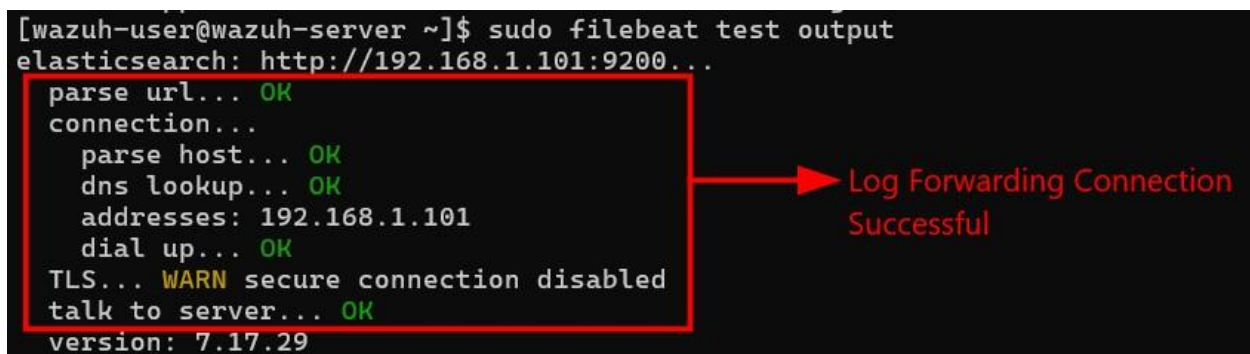
Objective	To verify that Filebeat is correctly running and forwarding logs to the ELK/Wazuh stack
Action	Check Filebeat service status and test output connection to ELK using Filebeat commands
Expected Result	Filebeat service is active and successfully connects to the ELK/OpenSearch output
Actual Result	Filebeat service was running and successfully established connection with the ELK/OpenSearch server, confirming log forwarding functionality
Conclusion	Test successful



```

• filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch.
   Loaded: loaded (/usr/lib/systemd/system/filebeat.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-04-18 02:41:58 UTC; 2 days ago
     Docs: https://www.elastic.co/products/beats/filebeat
    Main PID: 17456 (filebeat)
      Tasks: 7 (limit: 4633)
    Memory: 27.8M
       CPU: 22min 44.279s
    CGroup: /system.slice/filebeat.service
            └─17456 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebe
  
```

Figure 18 Filebeat service running on the Wazuh server, confirming log forwarding agent is active.



```

[wazuh-user@wazuh-server ~]$ sudo filebeat test output
elasticsearch: http://192.168.1.101:9200...
  parse url... OK
  connection...
    parse host... OK
    dns lookup... OK
    addresses: 192.168.1.101
    dial up... OK
  TLS... WARN secure connection disabled
  talk to server... OK
  version: 7.17.29
  
```

Log Forwarding Connection Successful

Figure 19 Filebeat successfully connected to the ELK/OpenSearch output, confirming log forwarding pipeline is operational.

Test Case 7

Unit Testing	Objectives
Objective	To verify that logs from the Windows endpoint are successfully ingested, indexed, and visualized in the ELK dashboard
Action	Generate system activity on the Windows endpoint and observe logs in the ELK Discover dashboard
Expected Result	Logs are indexed and displayed with key fields such as timestamp, agent information, and event data
Actual Result	A large volume of logs (4,893 hits) was successfully ingested and displayed in the ELK dashboard, showing event timeline distribution, timestamp, and source endpoint details
Conclusion	Test successful

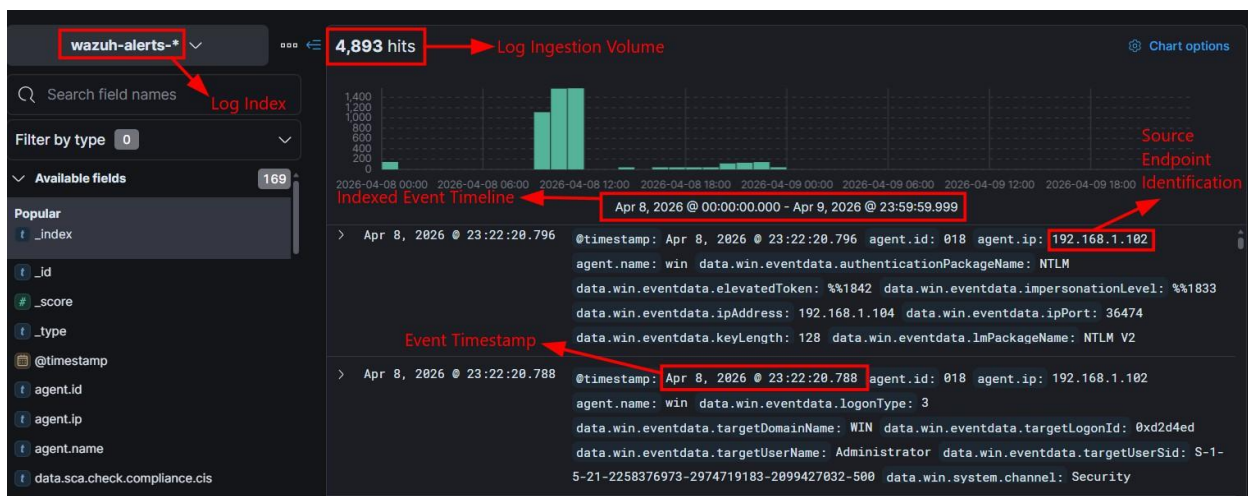


Figure 20 Logs from the Windows endpoint successfully indexed and visualized in the ELK dashboard, showing log ingestion volume, timeline distribution, and event metadata.

Test Case 8

Unit Testing	Objectives
Objective	To verify that multi-stage attack activities are detected and correctly classified using the custom rule set and MITRE ATT&CK mapping
Action	Perform a simulated attack from Kali Linux including reconnaissance, brute-force attempts, remote command execution, privilege escalation, account creation, account manipulation, and account deletion
Expected Result	Each stage of the attack is detected and mapped to the appropriate MITRE ATT&CK techniques
Actual Result	Multiple attack stages including reconnaissance, credential access, lateral movement, privilege escalation, execution, persistence, and impact were successfully detected and mapped to corresponding MITRE ATT&CK techniques
Conclusion	Test successful

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> net user testuser Pass123! /add
The command completed successfully.

PS C:\Windows\system32> net localgroup Administrators testuser /add
The command completed successfully.
```

Figure 21 Local user creation and administrative group assignment commands executed on the target system.


```
(kali@kali)-[~]
$ sudo nmap -sT -p 22,135,139,445 192.168.1.102
[sudo] password for kali:
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-20 07:57 EDT
Nmap scan report for 192.168.1.102
Host is up (0.069s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.49 seconds
```

Figure 22 Network reconnaissance scan identifying open services on the target host.

```
{
  "timestamp": "2026-04-20T03:40:43.164+0000",
  "rule": {
    "level": 8,
    "description": "Recon: Network service connection to port 22 from 192.168.57.10 using C:\\Windows\\System32\\OpenSSH\\sshd.exe",
    "id": "200011",
    "mitre": {
      "id": "T1046",
      "tactic": ["Discovery"],
      "technique": ["Network Service Discovery"]
    },
    "firedtimes": 1,
    "mail": true,
    "groups": ["attack_simulation_rules", "windows", "attack_chain"],
    "agent": {
      "id": "018",
      "name": "win",
      "ip": "192.168.1.102"
    },
    "manager": {
      "name": "wazuh-server",
      "id": "1776664733.665988",
      "full_log": {
        "win": {
          "system": {
            "providerName": "Microsoft-Windows-Security-Auditing",
            "providerGuid": "{54849625-5478-4994-a5ba-3e3...}"
          }
        }
      }
    }
  }
}
```

Figure 23 Reconnaissance activity detected (Rule 200011), mapped to T1046.

```
(kali@kali)-[~]
$ for i in {1..10}; do
  smbclient -L //192.168.1.102 -U admin%badpass > /dev/null 2>&1
done
```

Figure 24 Repeated authentication attempts simulating a brute-force attack against the target system.

```
s": "192.168.57.10",
"ipPort": "43958"
}},
"location": "EventChannel"
},
{
  "timestamp": "2026-04-20T05:58:53.131+0000",
  "rule": {
    "level": 8,
    "description": "Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10",
    "id": "200012",
    "mitre": {
      "id": "T1110",
      "tactic": ["Credential Access"],
      "technique": ["Brute Force"]
    },
    "firedtimes": 10,
    "mail": true,
    "groups": ["attack_simulation_rules", "windows", "attack_chain"],
    "agent": {
      "id": "018",
      "name": "win",
      "ip": "192.168.1.102"
    },
    "manager": {
      "name": "wazuh-server",
      "id": "1776664733.665988",
      "full_log": {
        "win": {
          "system": {
            "providerName": "Microsoft-Windows-Security-Auditing",
            "providerGuid": "{54849625-5478-4994-a5ba-3e3...}"
          }
        }
      }
    }
  }
}
```

Figure 25 Failed login attempts indicating brute-force attack (Rule 200012), mapped to T1110.

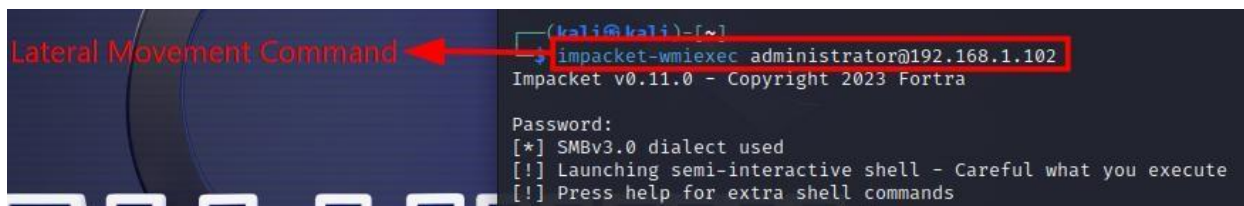


Figure 26 Remote command execution session initiated on the target system.

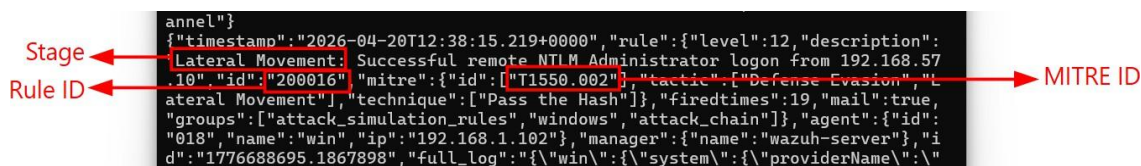


Figure 27 Successful remote NTLM login showing lateral movement (Rule 200016), mapped to T1550.002.

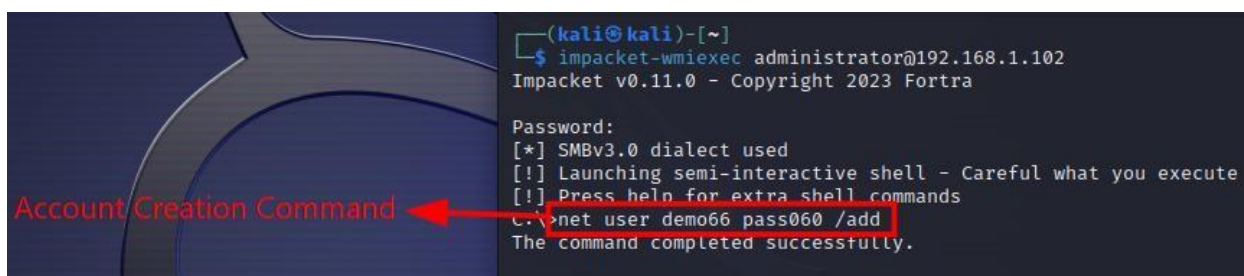


Figure 28 Command used to create a new local user account on the system.

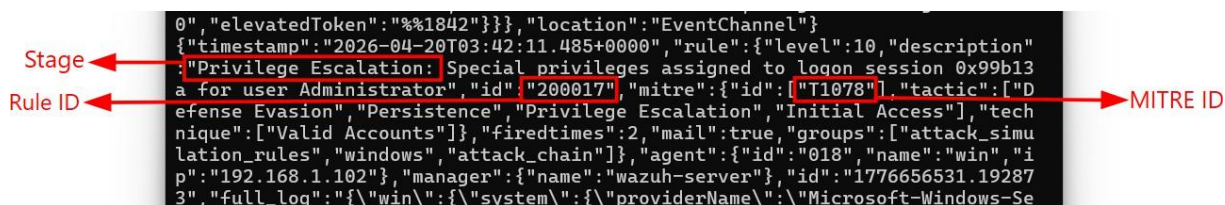


Figure 29 Elevated privileges assigned to Administrator session (Rule 200017), mapped to T1078.

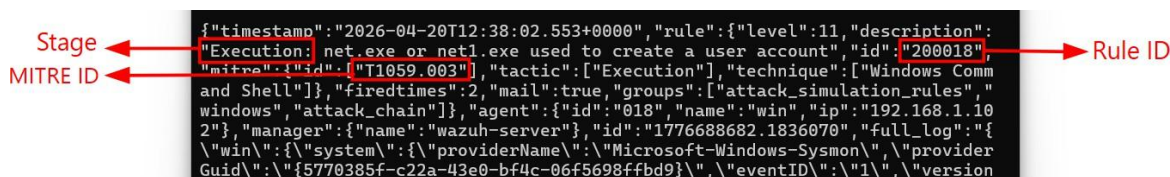


Figure 30 User account creation command detected (Rule 200018), mapped to T1059.003.

Stage → Rule ID → MITRE ID →

```

"location": "EventChannel"
{
  "timestamp": "2026-04-20T03:43:23.250+0000",
  "rule": {
    "level": 11,
    "description": "Persistence: Local user account created",
    "id": "200013",
    "mitre": {
      "id": "T1136.001",
      "tactic": ["Persistence"],
      "technique": ["Local Account"]
    },
    "fire": {
      "times": 2,
      "mail": true,
      "groups": ["attack_simulation_rules", "windows", "attack_chain"]
    },
    "agent": {
      "id": "018",
      "name": "win",
      "ip": "192.168.1.102"
    },
    "manager": {
      "name": "wazuh-server",
      "id": "1776656603.236996",
      "full_log": {
        "win": {
          "system": {
            "providerName": "Microsoft-Windows-Security-Auditing",
            "providerGuid": "54849625-5478-4994-a5ba-3e3b0328c30d",
            "eventID": "4732"
          }
        }
      }
    }
  }
}

```

Figure 31 Local user account creation establishing persistence (Rule 200013), mapped to T1136.001.

Privilege Escalation Command →

```

C:\>net localgroup Administrators demo66 /add
The command completed successfully.

```

Figure 32 Command used to add a user to the local Administrators group.

Stage → Rule ID → MITRE ID →

```

"location": "EventChannel"
{
  "timestamp": "2026-04-20T12:40:39.093+0000",
  "rule": {
    "level": 12,
    "description": "Privilege Escalation: net.exe or net1.exe used to add a user to Administrators",
    "id": "200019",
    "mitre": {
      "id": "T1098",
      "tactic": ["Persistence"],
      "technique": ["Account Manipulation"]
    },
    "fire": {
      "times": 2,
      "mail": true,
      "groups": ["attack_simulation_rules", "windows", "attack_chain"]
    },
    "agent": {
      "id": "018",
      "name": "win",
      "ip": "192.168.1.102"
    },
    "manager": {
      "name": "wazuh-server",
      "id": "1776688839.1898594",
      "full_log": {
        "win": {
          "system": {
            "providerName": "Microsoft-Windows-Sysmon",
            "providerGuid": "5770385f-c22a-43e0-bf4c-06f5698ffbd9",
            "eventID": "1"
          }
        }
      }
    }
  }
}

```

Figure 33 User added to the Administrators group via command activity (Rule 200019), mapped to T1098.

Stage → Rule ID → MITRE ID →

```

"location": "EventChannel"
{
  "timestamp": "2026-04-20T09:10:04.352+0000",
  "rule": {
    "level": 12,
    "description": "Privilege Escalation: User added to local Administrators group",
    "id": "200014",
    "mitre": {
      "id": "T1098",
      "tactic": ["Persistence"],
      "technique": ["Account Manipulation"]
    },
    "fire": {
      "times": 1,
      "mail": true,
      "groups": ["attack_simulation_rules", "windows", "attack_chain"]
    },
    "agent": {
      "id": "018",
      "name": "win",
      "ip": "192.168.1.102"
    },
    "manager": {
      "name": "wazuh-server",
      "id": "1776676204.779353",
      "full_log": {
        "win": {
          "system": {
            "providerName": "Microsoft-Windows-Security-Auditing",
            "providerGuid": "54849625-5478-4994-a5ba-3e3b0328c30d",
            "eventID": "4732"
          }
        }
      }
    }
  }
}

```

Figure 34 User addition to the Administrators group confirmed by system event (Rule 200014), mapped to T1098.

```

PS C:\Windows\system32> net user testuser /delete
The command completed successfully.
PS C:\Windows\system32>

```

Figure 35 Command used to delete a user account from the system.

Stage → Rule ID → MITRE ID →

```

"location": "EventChannel"
{
  "timestamp": "2026-04-20T12:16:11.575+0000",
  "rule": {
    "level": 10,
    "description": "Impact: User account deleted",
    "id": "200015",
    "mitre": {
      "id": "T1531",
      "tactic": ["Impact"],
      "technique": ["Account Access Removal"]
    },
    "fire": {
      "times": 1,
      "mail": true,
      "groups": ["attack_simulation_rules", "windows", "attack_chain"]
    },
    "agent": {
      "id": "018",
      "name": "win",
      "ip": "192.168.1.102"
    },
    "manager": {
      "name": "wazuh-server",
      "id": "1776687371.1769242",
      "full_log": {
        "win": {
          "system": {
            "providerName": "Microsoft-Windows-Security-Auditing",
            "providerGuid": "54849625-5478-4994-a5ba-3e3b0328c30d",
            "eventID": "4732"
          }
        }
      }
    }
  }
}

```

Figure 36 User account deletion showing impact (Rule 200015), mapped to T1531.

Test Case 9

Unit Testing	Objectives
Objective	To verify that critical event attributes such as attacker IP, target host, rule ID, event ID, attack stage, and session information are correctly extracted and structured from security events
Action	Generate attack activity and observe processed event output to validate extraction and labeling of key fields
Expected Result	Event attributes including attacker IP, host, rule ID, event ID, attack stage, and session identifiers are accurately parsed and displayed
Actual Result	The system successfully extracted and displayed key attributes such as attacker IP, target host, rule ID, event ID, attack stage, and session information from correlated events
Conclusion	Test successful

```

[EVENT - CORRELATED_HOST ACTIVITY]
Attacker   : 192.168.57.10
Host       : 192.168.1.102
Chain Key  : 192.168.57.10->192.168.1.102
Description: Privilege Escalation: Special privileges assigned to logon session 0xe8295a for user Administrator
Rule ID    : 200017 | Event ID: 4672 | Stage: Privilege Escalation
Logon ID   : 0xe8295a
ML BASE    : 0.8757
FINAL RISK : 0.9909
SESSION RISK: 0.9909
Signals    : privileged_logon
Verdict    : CRITICAL - likely successful compromise / remote execution
-----

```

Diagram illustrating the extraction of key attributes from the event data:

- Attacker IP Extraction: 192.168.57.10
- Target Host Identification: 192.168.1.102
- Attack Chain: 192.168.57.10->192.168.1.102
- Event Description Parsing: Privilege Escalation: Special privileges assigned to logon session 0xe8295a for user Administrator
- Rule & Event Mapping: Rule ID: 200017 | Event ID: 4672 | Stage: Privilege Escalation
- Session Identification: Logon ID: 0xe8295a

Figure 37 Parsed and correlated event data showing extraction of key attributes

Test Case 10

Unit Testing	Objectives
Objective	To verify that multiple related events are correlated using a common chain key
Action	Generate multi-stage attack activity and observe chain key across events
Expected Result	Events share the same chain key and are grouped into a single attack sequence
Actual Result	Events shared the same chain key, confirming successful correlation and sequence linking

Conclusion	Test successful
------------	-----------------

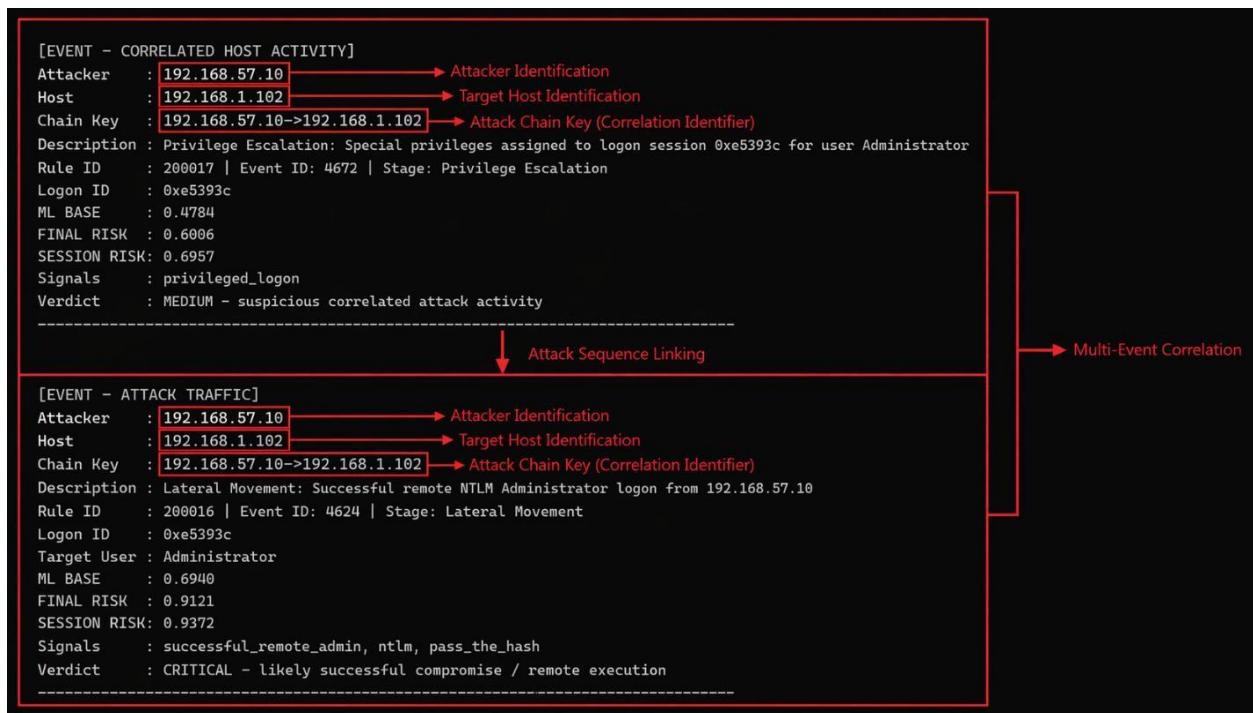


Figure 38 Correlated events showing shared chain key and attack progression from privilege escalation to lateral movement.

Test Case 11

Unit Testing	Objectives
Objective	To verify that ML BASE, FINAL RISK, and SESSION RISK are correctly generated and reflect event-level risk, contextual risk, and cumulative session risk respectively
Action	Generate attack activity (e.g., repeated failed login attempts) and observe the corresponding ML BASE, FINAL RISK, and SESSION RISK values

Expected Result	ML BASE reflects individual event risk, FINAL RISK incorporates contextual factors, and SESSION RISK increases with accumulation of correlated events
Actual Result	The event was assigned a moderate ML BASE score, FINAL RISK increased based on contextual evaluation, and SESSION RISK exceeded FINAL RISK, confirming cumulative risk aggregation across correlated events
Conclusion	Test successful

```

[EVENT - CORRELATED HOST ACTIVITY]
Attacker      : 192.168.57.10
Host          : 192.168.1.102
Chain Key     : 192.168.57.10->192.168.1.102
Description   : Credential Access: Failed remote network logon
                attempt for user admin from 192.168.57.10
Rule ID      : 200012 | Event ID: 4625 | Stage: Initial Access
Logon ID     : 0x0
Target User  : admin
ML BASE      : 0.6831
FINAL RISK   : 0.7414
SESSION RISK : 0.7759
Signals      : failed_login, ntlm
Verdict      : MEDIUM - suspicious correlated attack activity
-----

```

The diagram illustrates the risk scoring process. It starts with the ML BASE score of 0.6831, which is used for Contextual Risk Calculation to determine the FINAL RISK of 0.7414. This FINAL RISK is then used for Aggregated Session Risk to determine the SESSION RISK of 0.7759. The SESSION RISK is shown to be higher than the FINAL RISK, indicating cumulative risk aggregation.

Figure 39 Risk scoring showing ML BASE, FINAL RISK, and SESSION RISK, with SESSION RISK increasing due to correlated attack activity.

Test Case 12

Unit Testing	Objectives
Objective	To verify that normal system events are assigned low risk scores and do not trigger correlation or session-level risk escalation
Action	Observe standard host activity events such as user logoff and system/application events without performing any malicious actions
Expected Result	Events are classified as low risk with minimal ML BASE values, and FINAL RISK remains equal to SESSION RISK due to absence of correlation

Actual Result	The observed events were assigned low ML BASE, FINAL RISK, and SESSION RISK values, with FINAL RISK equal to SESSION RISK, confirming no correlation or accumulation
Conclusion	Test successful


```

[EVENT - HOST ACTIVITY]
Host      : 192.168.1.102
Chain Key : 192.168.1.102
Description : Windows User Logoff
Rule ID   : 60137 | Event ID: 4634 | Stage: Other
Logon ID  : 0xe828c1
Target User : Administrator
ML BASE   : 0.0015
FINAL RISK : 0.0152
SESSION RISK: 0.0152
Verdict   : LOW - weak or early-stage signal

```

→ Low Event-Level Risk (ML BASE)
→ Low Contextual Risk (FINAL RISK-No Correlation Detected)
→ Session Risk (Same as FINAL RISK-No Accumulation)
→ Low Verdict

```

[EVENT - HOST ACTIVITY]
Host      : 192.168.1.102
Chain Key : 192.168.1.102
Description : Summary event of the report's signatures.
Rule ID   : 60608 | Event ID: 1001 | Stage: Other
ML BASE   : 0.0000
FINAL RISK : 0.0175
SESSION RISK: 0.0175
Verdict   : LOW - weak or early-stage signal

```

```

[EVENT - HOST ACTIVITY]
Host      : 192.168.1.102
Chain Key : 192.168.1.102
Description : Windows application error event.
Rule ID   : 60602 | Event ID: 1002 | Stage: Other
ML BASE   : 0.2953
FINAL RISK : 0.2661
SESSION RISK: 0.2661
Verdict   : LOW - weak or early-stage signal

```

Figure 40 Normal activity with low-risk scores; FINAL RISK and SESSION RISK remain equal due to no correlation.

Test Case 13

Unit Testing	Objectives
Objective	To verify that the active response mechanism automatically blocks the attacker IP through pfSense when the calculated final risk exceeds the defined threshold
Action	Perform a high-risk simulated attack from Kali Linux, observe the generated Wazuh alert and risk score, and verify that the attacker IP is added to the pfSense block table and subsequent connections are denied

Expected Result	When the final risk score reaches or exceeds the response threshold, the attacker IP is automatically blocked in pfSense and further communication attempts fail
Actual Result	A high-risk lateral movement event produced a final risk score above the defined threshold, triggering the response mechanism. The attacker IP was added to the pfSense blocked_ips table, and further connection attempts from Kali Linux timed out successfully
Conclusion	Test successful

```

[EVENT - ATTACK TRAFFIC]
Attacker   : 192.168.57.10
Host       : 192.168.1.102
Chain Key  : 192.168.57.10->192.168.1.102
Description : Lateral Movement: Successful remote NTLM Administrator logon from 192.168.57.10
Rule ID    : 200016 | Event ID: 4624 | Stage: Lateral Movement → Attack Detection
Logon ID   : 0x10fba49
Target User : Administrator
ML BASE    : 0.6783
FINAL RISK : 0.9440
SESSION RISK: 0.9440
Signals    : successful_remote_admin, ntlm, pass_the_hash
Verdict    : CRITICAL - likely successful compromise / remote execution
[RESPONSE - TRIGGERED]
Time       : 2026-04-20 17:55:30.984847
Attacker   : 192.168.57.10
Host       : 192.168.1.102
Stage      : Lateral Movement
Final Risk : 0.9440 → Risk Threshold Exceeded (Auto-Block Trigger)
Verdict    : CRITICAL - likely successful compromise / remote execution
Action     : BLOCKED VIA PFSENSE + STATES KILLED → Automated Response

```

Figure 41 Detection of a lateral movement attack (Rule 200016) exceeding the risk threshold, triggering automated blocking of the attacker IP via pfSense.

```

(kali@kali)-[~]
$ impacket-wmiexec administrator@192.168.1.102

Impacket v0.11.0 - Copyright 2023 Fortra

Password:
[*] SMBv3.0 dialect used
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out

```

Figure 42: Connection attempts from the attacker fail after automated blocking, confirming that access has been successfully denied.

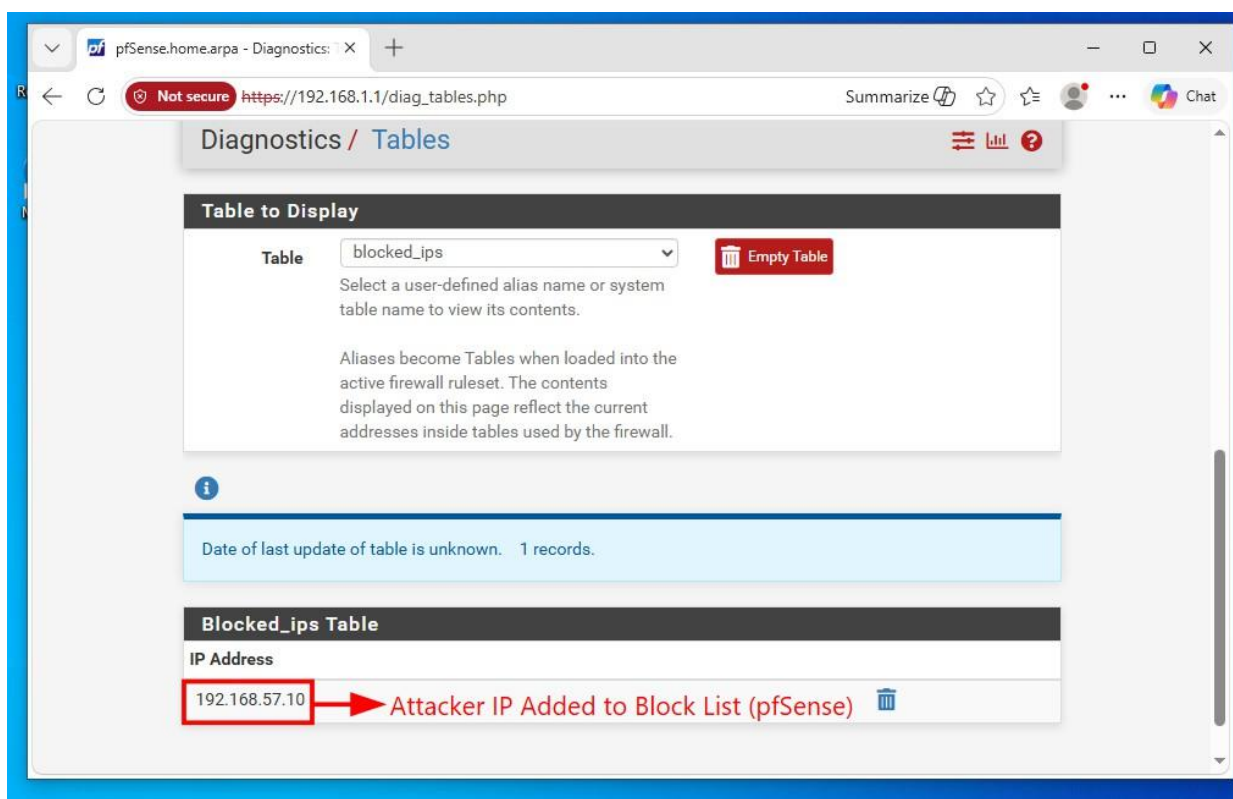


Figure 43 Attacker IP successfully added to the pfSense block table, confirming automated response enforcement.

Test Case 14

Unit Testing	Objectives
Objective	To verify that the automated response mechanism is triggered only when the defined risk threshold (≥ 0.90) is exceeded
Action	Perform multi-stage attack activity leading to increased risk and observe ML BASE, FINAL RISK, and SESSION RISK values
Expected Result	Risk values remain below the defined threshold (≥ 0.90), and no automated response is triggered
Actual Result	FINAL RISK and SESSION RISK remained below the threshold (0.90), and no automated response was triggered.
Conclusion	Test successful


```
[EVENT - CORRELATED HOST ACTIVITY]
Attacker   : 192.168.57.10
Host       : 192.168.1.102
Chain Key  : 192.168.57.10->192.168.1.102
Description: Privilege Escalation: Special privileges assigned to logon session 0xe65575 for user Administrator
Rule ID    : 200017 | Event ID: 4672 | Stage: Privilege Escalation
Logon ID   : 0xe65575
ML BASE    : 0.4569
FINAL RISK : 0.8853
SESSION RISK: 0.8991
Signals    : privileged logon
Verdict     : HIGH - likely lateral movement / privileged compromise
```

Annotations:

- ML BASE (0.4569) → Moderate Event-Level Risk
- FINAL RISK (0.8853) → High Contextual Risk (Near Response Threshold)
- SESSION RISK (0.8991) → Accumulated Session Risk (Near Threshold)
- Verdict (HIGH - likely lateral movement / privileged compromise) → High Risk Classification (Pre-Response State)

Figure 44 High-risk event showing ML BASE, FINAL RISK, and SESSION RISK approaching but remaining below the response threshold (0.90), with no automated blocking triggered.

Test Case 15

Unit Testing	Objectives
Objective	To verify that detected attack events are correctly visualized in the ELK dashboard, mapped to MITRE ATT&CK techniques, and that high-risk attackers are identified and blocked via pfSense
Action	Apply filters for custom rules in the ELK dashboard, observe event logs with rule IDs, MITRE mappings, and attack stages, and verify blocked IPs in the pfSense response table
Expected Result	Events are displayed with correct timestamps, rule IDs, MITRE ATT&CK mappings, and attack stage classifications,
	and high-risk attacker IPs are listed in the pfSense blocked table
Actual Result	Attack events were successfully visualized with custom rule IDs, MITRE ATT&CK mappings, and attack stage classifications, and the attacker IP was correctly blocked and recorded in the pfSense response table
Conclusion	Test successful

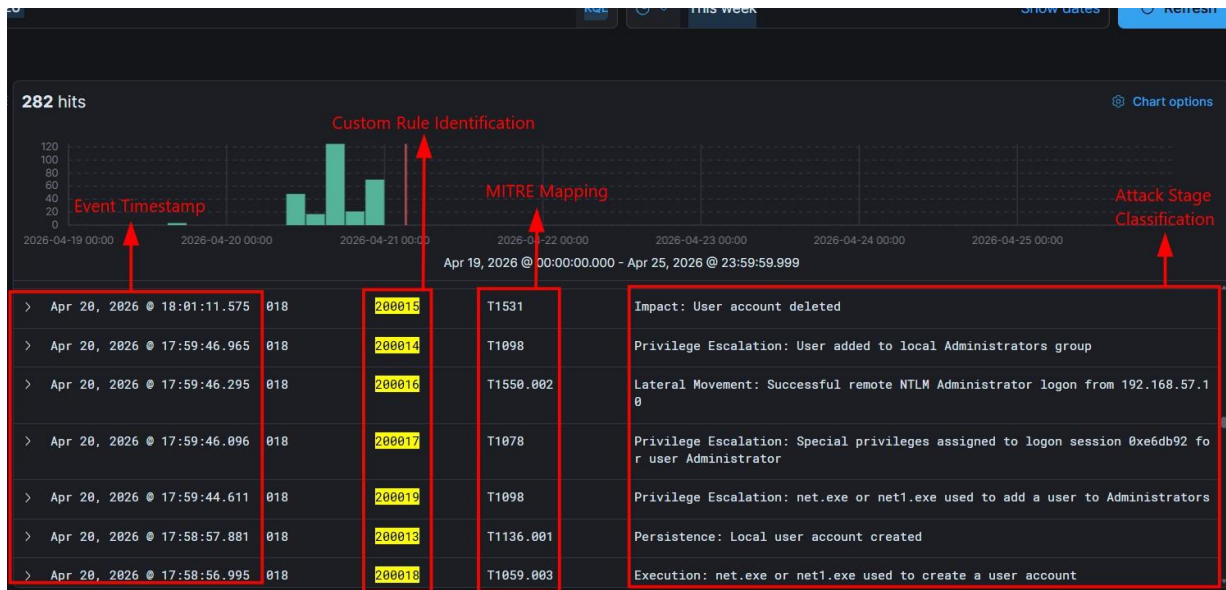


Figure 45 ELK dashboard visualizing Wazuh-detected events, including rule identification, MITRE ATT&CK mapping, and attack stage classification.

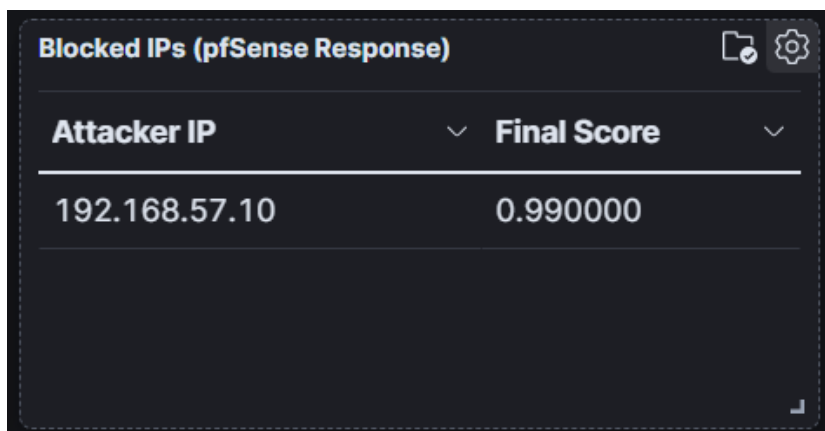


Figure 46 Automated response mechanism showing successful blocking of a high-risk attacker IP via pfSense integration.

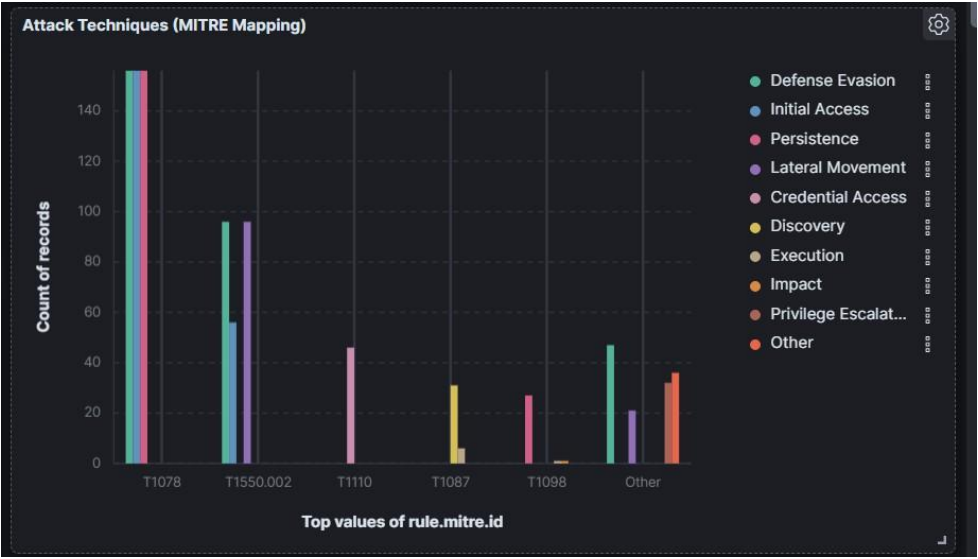


Figure 47 Graphical representation of detected attack techniques mapped to MITRE ATT&CK, highlighting attack stage distribution.